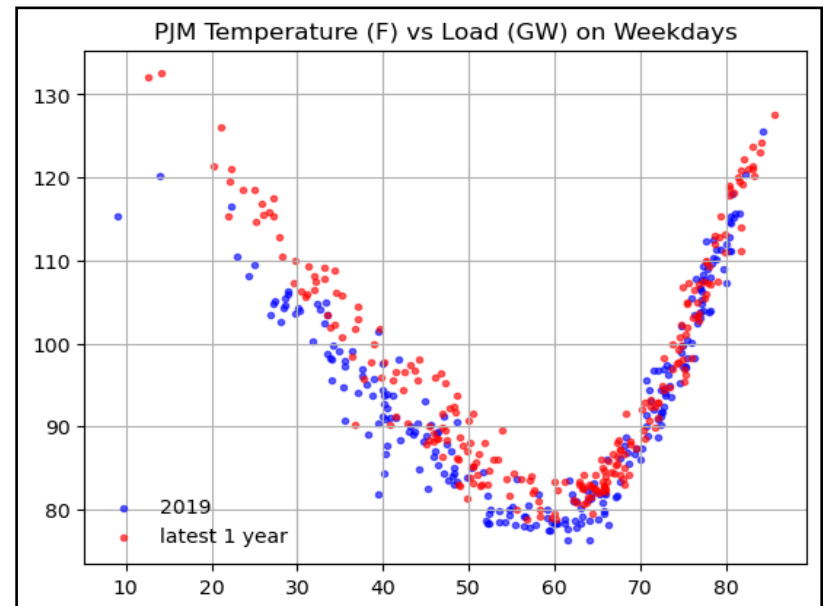
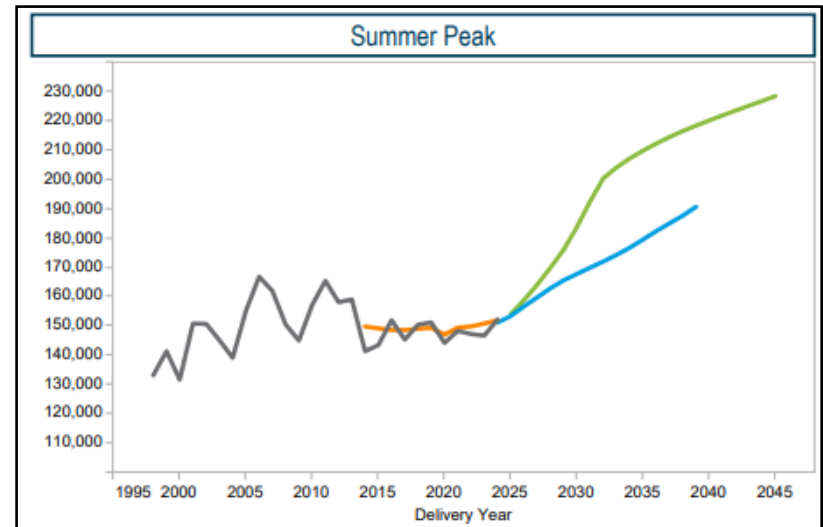

Energy Markets and Resource Adequacy: Unlocking Innovative Finance for Energy Solutions

Jason Lemme

Managing Director, Hartree Partners

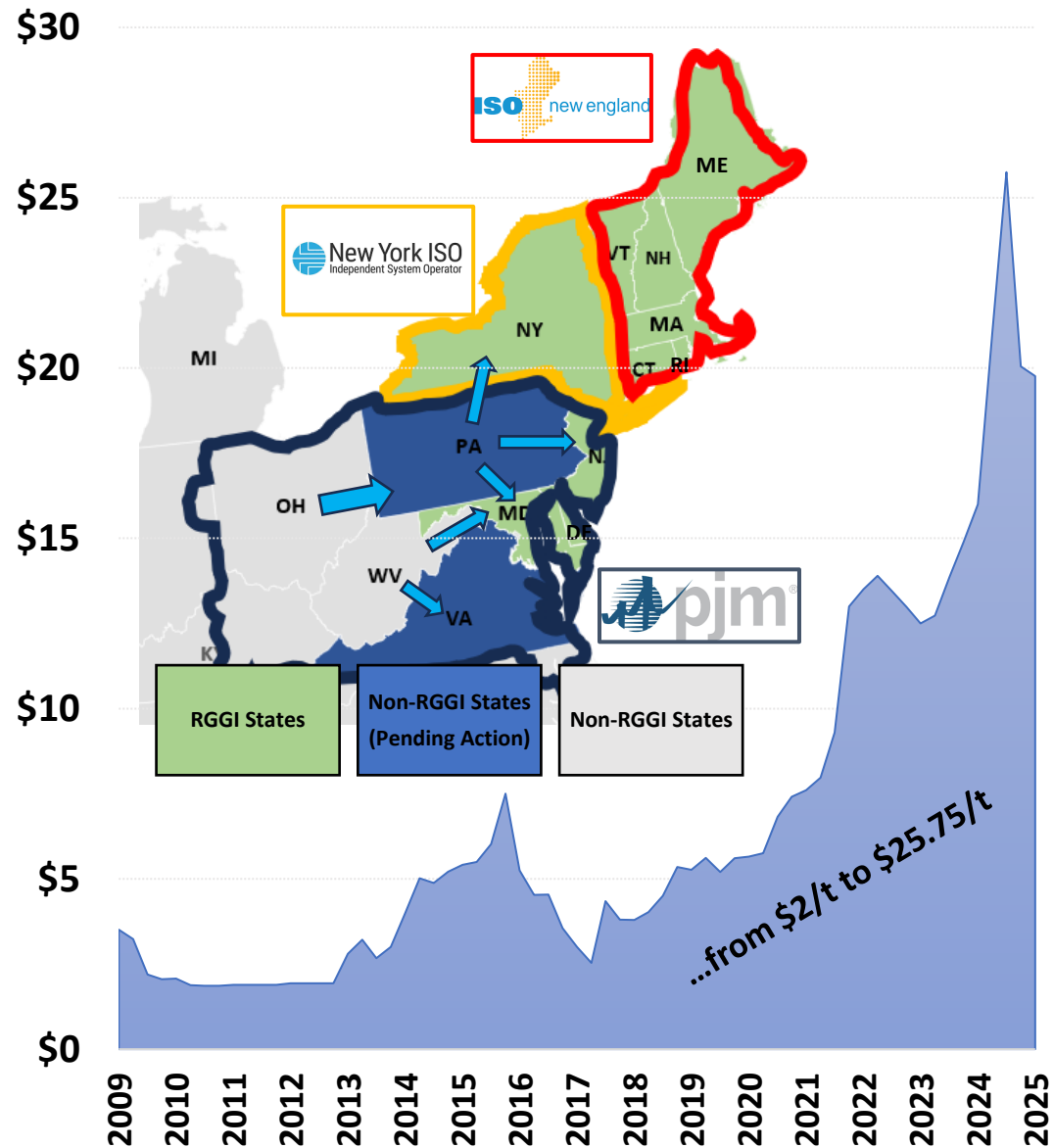
Unlocking Investment in New Jersey

- Current energy policies conceived in a 2005 world mindset
- What is different:
 - Demand is growing
(Electrification and Data centers)
 - Reliability costs are rising
(see Texas and California)
 - Higher winter needs
- Reliability is typically a summer issue, but the needs of winter are very different
- The policies that worked 15 years ago are doing the opposite of intended now and shipping economic growth to other states in PJM



Challenges of RGGI: Leakage and Weak Oversight

- Leakage: Shifts production to states with less stringent climate policies that results in higher emissions
 - RGGI does not address leakage and higher carbon pricing exacerbate the issue
 - Incentivizes out-of-region coal and less efficient gas that **increases** CO2 emissions
 - New Jersey ratepayers are subsidizing out-of-state generation that result in more global emissions
- Program has no ability to slow price increases since the “Cost Containment Reserve” has proven to be ineffective
- No Price Limit, No Position Limit, No Oversight



RGGI Leakage – More Non-RGGI Inefficient Gas Generation

Economics of NJ CCGT vs. PA Gas-Fired Generation

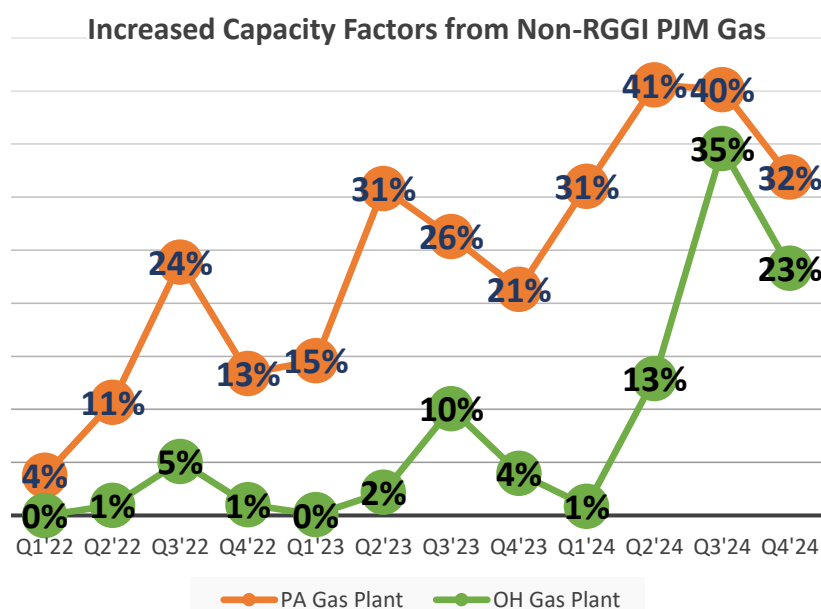
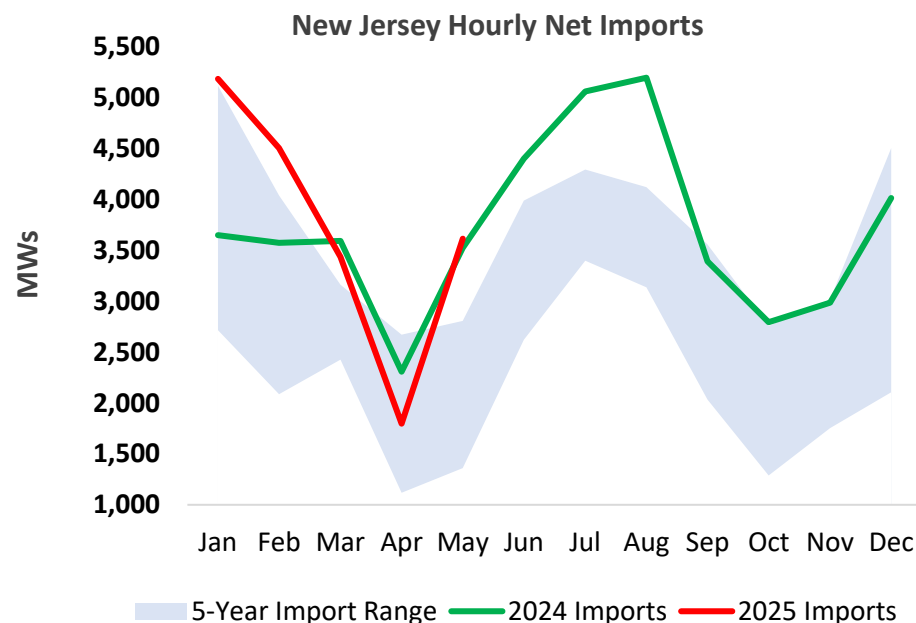
RGGI (\$/t)	% Higher Emission Rate of PA vs. NJ								
	0%	10%	20%	31%	41%	51%	61%	72%	82%
\$ -	0.24	1.38	2.52	3.66	4.80	5.94	7.08	8.22	9.36
\$ 2.00	(0.61)	0.52	1.66	2.80	3.94	5.08	6.22	7.36	8.50
\$ 4.00	(1.47)	(0.33)	0.81	1.95	3.09	4.22	5.36	6.50	7.64
\$ 6.00	(2.33)	(1.19)	(0.05)	1.09	2.23	3.37	4.51	5.65	6.79
\$ 8.00	(3.19)	(2.05)	(0.91)	0.23	1.37	2.51	3.65	4.79	5.93
\$ 10.00	(4.05)	(2.91)	(1.77)	(0.63)	0.51	1.65	2.79	3.93	5.07
\$ 12.00	(4.90)	(3.76)	(2.62)	(1.49)	(0.35)	0.79	1.93	3.07	4.21
\$ 14.00	(5.76)	(4.62)	(3.48)	(2.34)	(1.20)	(0.06)	1.08	2.21	3.35
\$ 16.00	(6.62)	(5.48)	(4.34)	(3.20)	(2.06)	(0.92)	0.22	1.36	2.50
\$ 18.00	(7.48)	(6.34)	(5.20)	(4.06)	(2.92)	(1.78)	(0.64)	0.50	1.64
\$ 20.00	(8.33)	(7.20)	(6.06)	(4.92)	(3.78)	(2.64)	(1.50)	(0.36)	0.78
\$ 22.00	(9.19)	(8.05)	(6.91)	(5.77)	(4.63)	(3.50)	(2.36)	(1.22)	(0.08)
\$ 24.00	(10.05)	(8.91)	(7.77)	(6.63)	(5.49)	(4.35)	(3.21)	(2.07)	(0.93)
\$ 27.50	(11.55)	(10.41)	(9.27)	(8.13)	(6.99)	(5.85)	(4.71)	(3.58)	(2.44)
\$ 28.00	(11.77)	(10.63)	(9.49)	(8.35)	(7.21)	(6.07)	(4.93)	(3.79)	(2.65)
\$ 30.00	(12.62)	(11.48)	(10.34)	(9.21)	(8.07)	(6.93)	(5.79)	(4.65)	(3.51)
\$ 32.00	(13.48)	(12.34)	(11.20)	(10.06)	(8.92)	(7.78)	(6.64)	(5.51)	(4.37)
\$ 34.00	(14.34)	(13.20)	(12.06)	(10.92)	(9.78)	(8.64)	(7.50)	(6.36)	(5.22)
\$ 36.00	(15.20)	(14.06)	(12.92)	(11.78)	(10.64)	(9.50)	(8.36)	(7.22)	(6.08)
\$ 38.00	(16.05)	(14.92)	(13.78)	(12.64)	(11.50)	(10.36)	(9.22)	(8.08)	(6.94)
\$ 40.00	(16.91)	(15.77)	(14.63)	(13.49)	(12.35)	(11.22)	(10.08)	(8.94)	(7.80)
\$ 42.00	(17.77)	(16.63)	(15.49)	(14.35)	(13.21)	(12.07)	(10.93)	(9.79)	(8.65)

In-State Gas Wins

Out-of-State Gas Wins

When RGGI prices are higher than \$10/t, a 41% less efficient Pennsylvania natural gas-fired facility with combustion turbines (10.5 Heat Rate) becomes more economically viable than an average New Jersey gas generator (7.30 Heat Rate)

More Imports, Additional Costs and Higher Global Emissions



	Location	Heat Rate Average	Capacity Factor Average
2024	New Jersey	7.30	32.89%
	Pennsylvania	8.39	55.06%
	OH Gas Plant	12.17	13% → 36%
	PA Gas Plant	10.87	4% → 18%

RGGI adds at least...

- 2.7 million short tons of annual system-wide CO2 emissions
- \$580 million in annual cost for New Jersey ratepayers with majority going to out-of-state generators

[Data Miner 2 - Hourly Net Exports by State \(pjm.com\)](https://pjm.com/data-miner/2-hourly-net-exports-by-state)

<https://campd.epa.gov/data/custom-data-download>

[https://tcr-us.com/uploads/3/5/9/1/35917440/tcr white paper rggi 2025 03 06.pdf](https://tcr-us.com/uploads/3/5/9/1/35917440/tcr_white_paper_rggi_2025_03_06.pdf)

Keys to New Investment and Reliability

- Form effective and realistic policy around growing winter reliability → that's where the future risk lies
- Pennsylvania and Ohio are beating New Jersey and not by a little bit
- Claim victory on RGGI and end it. It did a great job in eliminating coal plants in the RGGI states. But now you can lower bills and reduce global emissions in one move